

Gradient Test Sequence

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

2 Overview

The gradient test sequence is designed to test the gradient amplifiers as well as to run the NEMA dB/dt test. This protocol is only available in Research and Service mode. The **Gradtest** pulse sequence outputs a positive lobe trapezoid gradient immediately followed by a negative lobe trapezoid gradient. There are no RF transmit pulses or data acquisition periods. The gradient waveform characteristics such as duration, amplitude and gradient channel(s) are under the control of the user. Note that there are software and hardware limits on gradient slew rate and gradient amplitude. Below is a summary of the pulse sequence controls (refer to Figure 1).

<i>TR</i>	time in msec between the start of two consecutive gradient pulses
<i>TE</i>	duration in usec of the entire bipolar gradient pulse
<i>Freq</i>	ramp duration in usec
<i>Phase</i>	gradient amplitude in 1/100 G/cm (e.g. Phase=100=1.00 G/cm)

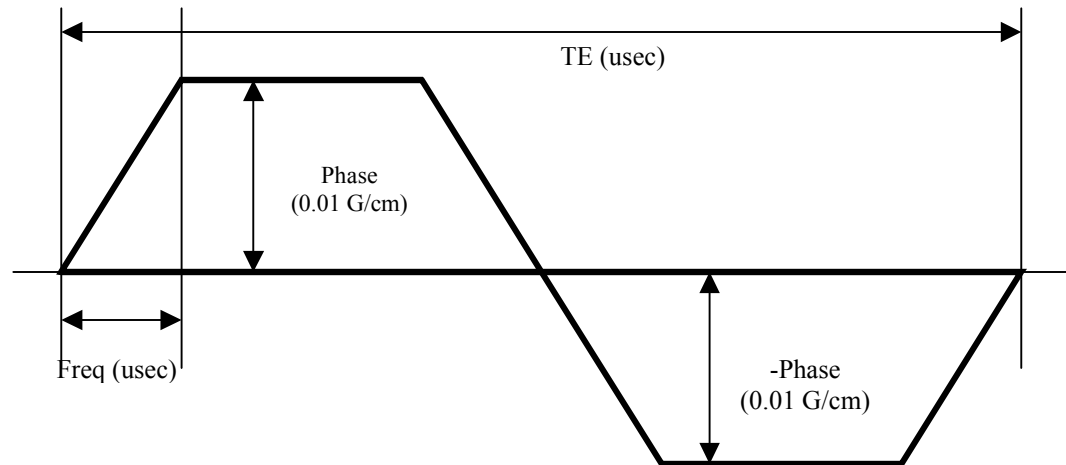


Figure 1: Grad test diagram

In addition to the scan parameters, the following scan options allow control over which gradient channel (Readout, Phase Encoding, or Slice Select) plays out the waveform. When the scan option is selected, the waveform is applied to the designated channel. More than one channel may be on at one time.

<i>RF Spoiling</i>	Readout channel	(X gradient if orientation is Axial on the prescription page)
<i>Partial Data</i>	Phase Encoding channel	(Y gradient if orientation is Axial on the prescription page)
<i>No Phase Wrap</i>	Slice Select channel	(Z gradient if orientation is Axial on the prescription page)

The following conditions must be met to run the sequence:

- The duration of TE must be larger than four times the ramp duration.
- The duration of TR must be greater than the duration of TE.

NOTICE This protocol must only be run in **prescan RG mode**.

3 Preliminary Requirements

3.1 Tools and Test Equipment

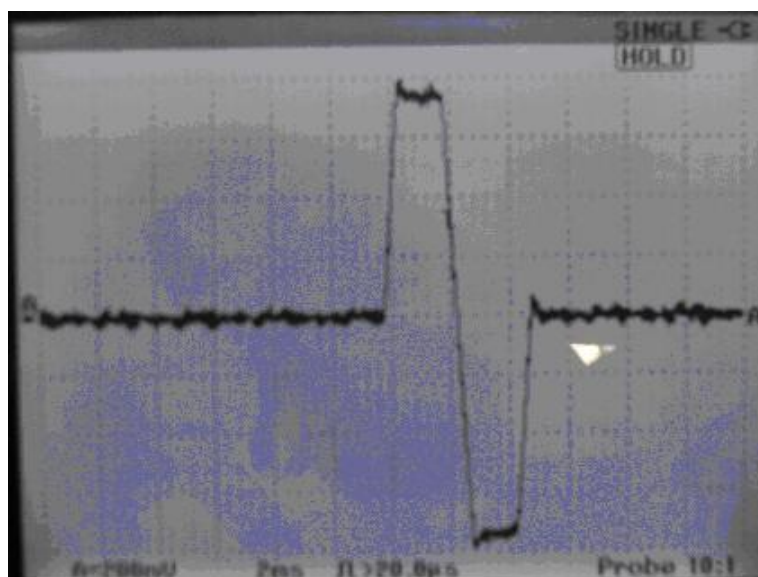
Item	Qty	Effectivity	Part#	Manufacturer
Oscilloscope	1			
Service Methods DVD	1	-	5000-9311-001	-

4 Procedure

4.1 Gradient Test

1. Remove the gradient amplifier protective cover.
2. Connect the oscilloscope to the **current monitor** connector located on the gradient amplifier being tested.
3. Go into **service mode**.
4. On the Scan Setup screen, select **Single Scan** and **List All**, and then select **GradTest** from the list of sequences.
5. Setup the pulse sequence controls (refer to Figure 1)
 - a. Frequency (ramp time) = 500
 - b. Phase (gradient strength) = 100
 - c. TE (total pulse duration) = 5000
 - d. TR = 1000
 - e. RF Spoiling option checked turns on the X gradient, Partial Data option checked turns on the Y gradient, and No Phase Wrap option checked turns on the Z gradient.
 - f. Select manual prescan
6. Select **Begin Protocol**
 - a. In the Manual Prescan window, click **RG mode**
 - b. Observe the waveform on the oscilloscope. See Figure 2

Figure 2: Typical Oscilloscope Waveform



5 Finalization

1. Disconnect the oscilloscope
2. Replace the gradient amplifier protective cover